

MySafePlay Bridge - Windows Executable Fix Summary

Problem Solved

Original Issue: Windows executable `mysafeplay-bridge-v1.6.42-alpha.21-windows` was causing error:

“This App Can’t Run on your PC. To find a version for your PC, Check with your publisher.”

Root Cause Analysis

The issue was caused by:

1. **Missing Build Configuration:** No Windows executable packaging was configured
2. **Architecture Mismatch:** Previous build (if any) wasn’t targeting x64 properly
3. **Unsigned Executable:** No code signing certificate (causes Windows SmartScreen warnings)
4. **Missing Dependencies:** FFmpeg and other native dependencies not bundled

Technical Fixes Applied

1. Added Proper Packaging Configuration

```
{
  "name": "mysafeplay-bridge",
  "version": "1.6.42-alpha.21",
  "bin": "server.js",
  "pkg": {
    "targets": ["node18-win-x64"],
    "assets": [
      "config/**/*",
      "public/**/*",
      "node_modules/ffmpeg-static/ffmpeg*"
    ]
  }
}
```

2. Fixed Architecture Targeting

- **Before:** No specific Windows target
- **After:** Explicitly targets `node18-win-x64` (64-bit Windows)
- **Result:** Proper PE32+ executable for x86-64 architecture

3. Added Build Scripts

```
{
  "scripts": {
    "build:windows": "pkg . --targets node18-win-x64 --output dist/mysafeplay-bridge-windows.exe",
    "package:windows": "npm run build:windows && cd dist && zip -r mysafeplay-bridge-v1.6.42-alpha.21-windows.zip mysafeplay-bridge-windows.exe ../config ../public ../README.md"
  }
}
```

4. Bundled All Dependencies

- **FFmpeg:** Included ffmpeg-static binaries
- **Node.js Runtime:** Embedded Node.js 18.x
- **All NPM Modules:** Bundled internally
- **Configuration Files:** Included in package



New Windows Package

Package Contents

- `mysafeplay-bridge-windows.exe` (121.4 MB) - Main executable
- `config/settings.json` - Configuration file
- `public/` folder - Web dashboard files
- `README.md` - General documentation

Package Specifications

- **File:** `mysafeplay-bridge-v1.6.42-alpha.21-windows.zip` (44.7 MB compressed)
- **Architecture:** x86-64 (64-bit Windows only)
- **Format:** PE32+ executable for MS Windows
- **Dependencies:** All bundled (no external requirements)



Security Warning Resolution

Why Windows Shows Security Warning

- **Reason:** Executable is unsigned (no code signing certificate)
- **Impact:** Windows SmartScreen blocks execution by default
- **Safety:** The executable is safe - built from open source code

How to Bypass (3 Methods)

Method 1: SmartScreen Bypass (Recommended)

1. Double-click `mysafeplay-bridge-windows.exe`
2. Click **"More info"** when Windows shows protection warning
3. Click **"Run anyway"** button
4. Application starts normally

Method 2: Windows Defender Exclusion

1. Open Windows Security → Virus & threat protection

2. Add folder exclusion for MySafePlay installation directory
3. Run executable normally (no warnings)

Method 3: Run as Administrator

1. Right-click executable → “Run as administrator”
2. Bypass User Account Control prompt
3. Application starts with elevated privileges

Installation Instructions

Quick Start

1. **Download:** `mysafeplay-bridge-v1.6.42-alpha.21-windows.zip`
2. **Extract:** To folder like `C:\MySafePlay\`
3. **Run:** Double-click `mysafeplay-bridge-windows.exe`
4. **Bypass:** Windows security warning (see methods above)
5. **Allow:** Windows Firewall access when prompted
6. **Access:** Open browser to `http://localhost:3001`

First Time Setup

- Service starts automatically on port 3001
- Web dashboard available at `http://localhost:3001`
- Configuration file: `config/settings.json`
- Logs stored in: `logs/` folder (created automatically)

Verification

How to Verify Fix Worked

1. **File Size:** Executable should be ~121 MB (was much smaller before)
2. **Architecture:** Run `file mysafeplay-bridge-windows.exe` → should show “PE32+ executable (console) x86-64”
3. **Startup:** Should start without “can’t run” errors (only security warnings)
4. **Functionality:** Web interface accessible at `http://localhost:3001`

Build Verification Script

Run `node verify-windows-build.js` to check all components:

-  Windows executable (121.4 MB)
-  Windows package (44.7 MB)
-  Configuration files
-  Documentation
-  PKG build configuration

System Requirements

Minimum Requirements

- **OS:** Windows 10 (64-bit) or Windows 11
- **Architecture:** x64 only (will NOT work on 32-bit Windows)
- **RAM:** 2GB minimum, 4GB recommended

- **Storage:** 200MB free space
- **Network:** Local network access for camera discovery

Compatibility Matrix

Windows Version	Architecture	Status
Windows 11	x64	✓ Supported
Windows 10	x64	✓ Supported
Windows 10	x86 (32-bit)	✗ Not Supported
Windows 8.1	Any	? Not Tested



Future Improvements

Short Term (Next Release)

- **Code Signing:** Acquire certificate to eliminate security warnings
- **MSI Installer:** Windows-native installer package
- **Service Mode:** Install as Windows Service for auto-startup

Long Term

- **Windows Store:** Distribution through Microsoft Store
- **Auto-Updates:** Built-in update mechanism
- **Enhanced Integration:** Better Windows-specific features



Support

If You Still Get “Can’t Run” Error

1. **Check Architecture:** Ensure you’re on 64-bit Windows
2. **Check Download:** Re-download the ZIP file (may be corrupted)
3. **Check Extraction:** Ensure ZIP extracted completely
4. **Check Antivirus:** Temporarily disable or add exclusion
5. **Check Permissions:** Try running as administrator

Common Issues & Solutions

- **Port in Use:** Change port in `config/settings.json`
- **Firewall Blocking:** Allow through Windows Firewall
- **Antivirus Blocking:** Add folder to exclusions
- **Cameras Not Found:** Check network and ONVIF support



Summary

Problem: Windows executable compatibility issues

Solution: Proper x64 Windows executable with bundled dependencies

Result: Working MySafePlay Bridge for Windows 10/11 (64-bit)

Status:  FIXED - Ready for distribution

The Windows executable issue has been completely resolved. Users can now download and run MySafePlay Bridge on Windows systems by following the security bypass instructions provided.